Kähler Geometry

<u>Definition</u> A <u>complex manifold</u> M is a smooth manifold with a cover $M = \bigcup_\alpha U_\alpha$ together with homeomorphisms $\varphi_\alpha : U_\alpha \to V_\alpha \subset \mathbb{C}^n$ such that transition maps

$\varphi_\alpha \circ \varphi_\beta^{-1}$ are holomorphic on their domains.

φ_α defines local holomorphic coordinates on U_α

<u>E.g.</u> Complex projective space $\mathbb{CP}^n$ is quotient space $(\mathbb{C}^{n+1} \setminus \{0\}) / \sim$

where $(z_0, \dots, z_n) \sim (\lambda z_0, \dots, \lambda z_n)$ for $\lambda \in \mathbb{C}$

denote equiv. class by $[z_0 : \dots : z_n]$

$z_0, \dots, z_n$ are homogeneous coordinates

$$\mathbb{CP}^n = \bigcup_{i=0}^{n} U_i$$

$$U_i = \{ [z_0 : \dots : z_n] \mid z_i \neq 0 \}$$

$$\varphi_i : U_i \longrightarrow \mathbb{C}^n$$

$$[z_0 : \dots : z_n] \longmapsto \left(\frac{z_0}{z_i}, \dots, \overset{\text{omit}}{\widehat{\frac{z_i}{z_i}}}, \dots \frac{z_n}{z_i} \right)$$

$\mathbb{CP}^n$ is compact. $\mathbb{CP}^n$ is homeomorphic to S^{2n+1}/S^1

<u>E.g.</u> Projective manifolds

If $f_1, \dots, f_k$ are homogeneous polynomials in $z_0, \dots, z_n$, then their common zero set

$$V = \left\{ [z_0 : \dots : z_n] \;\middle|\; \begin{array}{l} f_i(z_0, \dots, z_n) = 0 \\ \text{for } i = 1, \dots, k \end{array} \right\} \subset \mathbb{CP}^n$$

is well defined.

If V is smooth then it is a complex manifold.

<u>Definition</u> An <u>almost complex structure</u> on a smooth manifold M is

$$J : TM \longrightarrow TM$$

such that $J^2 = -\mathrm{Id}$.

E.g. On a complex manifold if $z^1, \dots, z^n$ are local holomorphic coords and $z^i = x^i + \sqrt{-1}\, y^i$, let

$$J\left(\frac{\partial}{\partial x^i}\right) = \frac{\partial}{\partial y^i}, \qquad J\left(\frac{\partial}{\partial y^i}\right) = -\frac{\partial}{\partial x^i}$$

transition maps are holomorphic $\Longrightarrow$ J is well defined

Complexified tangent and cotangent bundles

$$T^{\mathbb{C}}M = TM \otimes_{\mathbb{R}} \mathbb{C} \qquad\qquad \Omega^1_{\mathbb{C}} M = \Omega^1 M \otimes_{\mathbb{R}} \mathbb{C}$$

local bases $\left\{ \frac{\partial}{\partial z^1}, \dots, \frac{\partial}{\partial z^n}, \frac{\partial}{\partial \bar{z}^1}, \dots, \frac{\partial}{\partial \bar{z}^n} \right\}$ $\qquad \left\{ dz^1, \dots, dz^n, d\bar{z}^1, \dots, d\bar{z}^n \right\}$

$$\frac{\partial}{\partial z^i} = \frac{1}{2}\left(\frac{\partial}{\partial x^i} - \sqrt{-1}\frac{\partial}{\partial y^i}\right), \quad \frac{\partial}{\partial \bar{z}^i} = \frac{1}{2}\left(\frac{\partial}{\partial x^i} + \sqrt{-1}\frac{\partial}{\partial y^i}\right) \qquad \begin{aligned} dz^i &= dx^i + \sqrt{-1}\, dy^i \\ d\bar{z}^i &= dx^i - \sqrt{-1}\, dy^i \end{aligned}$$

$$T^{\mathbb{C}}M = T^{1,0}M \oplus T^{0,1}M \qquad \Omega^1_{\mathbb{C}} = \Omega^{1,0}M \oplus \Omega^{0,1}M$$

$\mathrm{span}\left\{\frac{\partial}{\partial z^i}\right\}$ — $\sqrt{-1}$ eigenspace of J

$\mathrm{span}\left\{\frac{\partial}{\partial \bar{z}^i}\right\}$ — $-\sqrt{-1}$ eigenspace of J

higher degree forms $\Omega^r_{\mathbb{C}}M = \bigoplus_{p+q=r} \Omega^{p,q}M$

(p,q)-forms $\Omega^{p,q}M$ spanned by

$$dz^{i_1} \wedge \cdots \wedge dz^{i_p} \wedge d\bar{z}^{j_1} \wedge \cdots \wedge d\bar{z}^{j_q}$$

exterior derivative decomposes as $d = \partial + \bar{\partial}$

$$\partial : \Omega^{p,q}M \longrightarrow \Omega^{p+1,q}M$$
$$\bar{\partial} : \Omega^{p,q}M \longrightarrow \Omega^{p,q+1}M$$

$$d^2 = 0 \iff \partial^2 = 0, \quad \bar{\partial}^2 = 0$$
$$\partial\bar{\partial} + \bar{\partial}\partial = 0$$

<u>E.g.</u> 1. $f: M \to \mathbb{C}$ is holomorphic

if and only if $\bar{\partial} f = 0$

2. For $f: M \to \mathbb{R}$

$\sqrt{-1}\, \partial\bar{\partial} f$ is a real $(1,1)$-form

If $f: \mathbb{C} \to \mathbb{R}$,

$$\sqrt{-1}\, \partial\bar{\partial} f = \frac{1}{2}\left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}\right) dx \wedge dy$$

$$\overline{\sqrt{-1}\, \partial\bar{\partial} f} = -\sqrt{-1}\, \bar{\partial}\partial \bar{f} \implies \sqrt{-1}\, \partial\bar{\partial} f \text{ is real}$$
$$= \sqrt{-1}\, \partial\bar{\partial} f$$

$$\sqrt{-1}\, \partial\bar{\partial} f = \sqrt{-1}\, \frac{\partial}{\partial z}\frac{\partial f}{\partial \bar{z}}\, dz \wedge d\bar{z}$$
$$= \sqrt{-1}\, \frac{1}{4}\left(\frac{\partial}{\partial x} - \sqrt{-1}\frac{\partial}{\partial y}\right)\left(\frac{\partial f}{\partial x} + \sqrt{-1}\frac{\partial f}{\partial y}\right)(dx + \sqrt{-1}\, dy) \wedge (dx - \sqrt{-1}\, dy)$$
$$= \sqrt{-1}\, \frac{1}{4}\left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}\right)(-\sqrt{-1}\, dx \wedge dy + \sqrt{-1}\, dy \wedge dx)$$
$$= \frac{1}{2}\left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}\right) dx \wedge dy$$

Hermitian and Kähler metrics

M complex manifold with complex structure J.

<u>Definition</u> 1. A Riemannian metric g on M is <u>Hermitian</u> if $g(JX, JY) = g(X, Y)$ for all X, Y.

2. If g is Hermitian its associated 2-form is

$$\omega(X, Y) = g(JX, Y)$$

ω is a real $(1,1)$-form.

g is <u>Kähler</u> if $d\omega = 0$.

In local coordinates a Hermitian g is determined by components

$$g_{j\bar{k}} = g\left(\frac{\partial}{\partial z^j}, \frac{\partial}{\partial \bar{z}^k}\right)$$

Since $\quad g\left(\frac{\partial}{\partial z^j}, \frac{\partial}{\partial z^k}\right) = g\left(\frac{\partial}{\partial \bar{z}^j}, \frac{\partial}{\partial \bar{z}^k}\right) = 0$

$g_{j\bar{k}}$ defines a positive definite Hermitian matrix at each point:

$$\overline{g_{j\bar{k}}} = g_{k\bar{j}}$$

and if $(\xi_1, \dots, \xi_n) \in \mathbb{C}^n \setminus \{0\}$

then
$$\sum_{j,k} g_{j\bar{k}} \xi_j \bar{\xi}_k > 0.$$

Associated 2-form is
$$\omega = \sqrt{-1} \sum_{j,k} g_{j\bar{k}} \, dz^j \wedge d\bar{z}^k$$

$$\omega\left(\frac{\partial}{\partial z^j}, \frac{\partial}{\partial \bar{z}^k}\right) = g\left(J\frac{\partial}{\partial z^j}, \frac{\partial}{\partial \bar{z}^k}\right) = g\left(\sqrt{-1}\frac{\partial}{\partial z^j}, \frac{\partial}{\partial \bar{z}^k}\right) = \sqrt{-1}\, g_{j\bar{k}}$$

g is Kähler if $\frac{\partial}{\partial z^i} g_{j\bar{k}} = \frac{\partial}{\partial z^j} g_{i\bar{k}}$

<u>E.g.</u> (Fubini-Study metric)

Recall projection map $\pi : \mathbb{C}^{n+1} \setminus \{0\} \longrightarrow \mathbb{CP}^n$

If $U \subset \mathbb{CP}^n$ is open we define a <u>section over U</u> to be a holomorphic

$$s : U \longrightarrow \mathbb{C}^{n+1} \setminus \{0\}$$

such that $\pi \circ s = \mathrm{id}$

Given a section s define

$$\omega_{FS} = \sqrt{-1}\,\partial\bar\partial \log \|s\|^2$$

This is well defined on $\mathbb{CP}^n$:

if s' is another section over V

then $s' = fs$ on $U \cap V$ for $f : U \cap V \to \mathbb{C} \setminus \{0\}$ holomorphic

so $\sqrt{-1}\,\partial\bar\partial \log \|s'\|^2 = \sqrt{-1}\,\partial\bar\partial \log |f|^2 \|s\|^2$

$$= \sqrt{-1}\,\partial\bar\partial \log \|s\|^2 + \sqrt{-1}\,\partial\bar\partial \log |f|^2$$

$\log |f|^2 = \log f + \log \bar f$

$$= \sqrt{-1}\,\partial\bar\partial \log \|s\|^2 .$$

To check that the components of ω_{FS} define positive definite matrices

note that ω_{FS} is $U(n+1)$ invariant

$\Longrightarrow$ enough to work at one point

at $[1 : 0 : \dots : 0]$ use local coordinates

$$w^i = \frac{z_i}{z_0} \quad \text{for } i = 1, \dots, n$$

and section $s(w^1, \dots, w^n) = (1, w^1, \dots, w^n)$ over U_0

$$\omega_{FS} = \sqrt{-1}\,\partial\bar\partial \log (1 + |w^1|^2 + \dots + |w^n|^2)$$

at origin $\omega_{FS} = \sqrt{-1} \sum_{i=1}^{n} dw^i \wedge d\bar w^i$